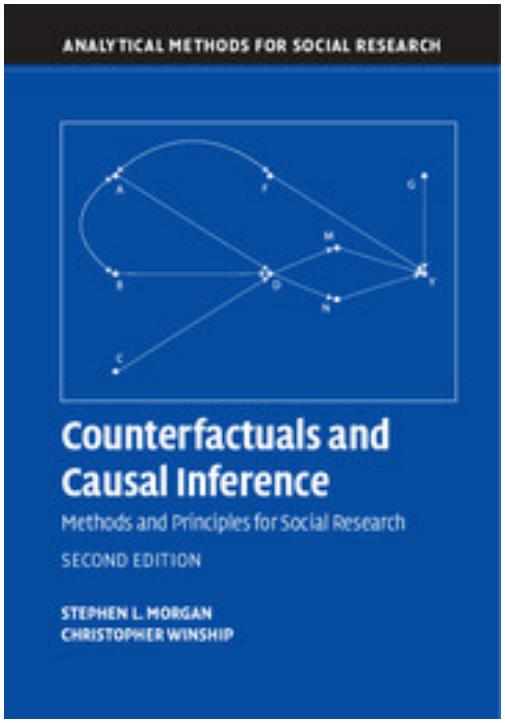
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 10

Outline

1. The Dangers of Insufficiently Deep Explanations
2. The Front-Door Criterion and and Identification of Causal Effects by Mechanism
3. The Appeal for Generative Mechanisms
4. The Pursuit of Explanation with Mechanisms That Bottom Out

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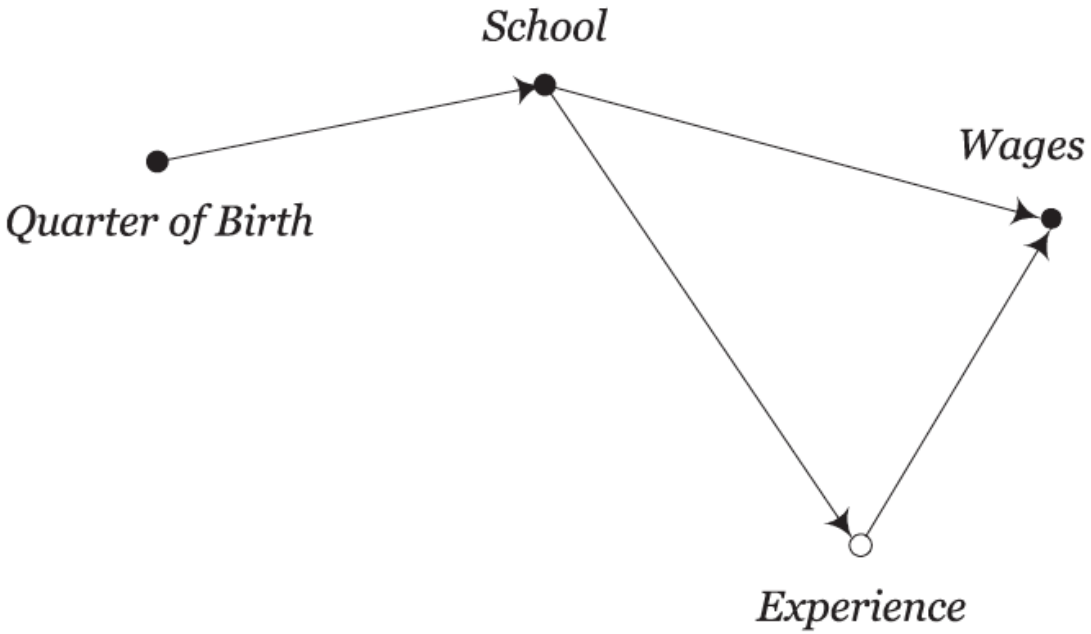

Figure 10.1 A directed graph for compliers with quarter of birth as an IV for years of schooling.

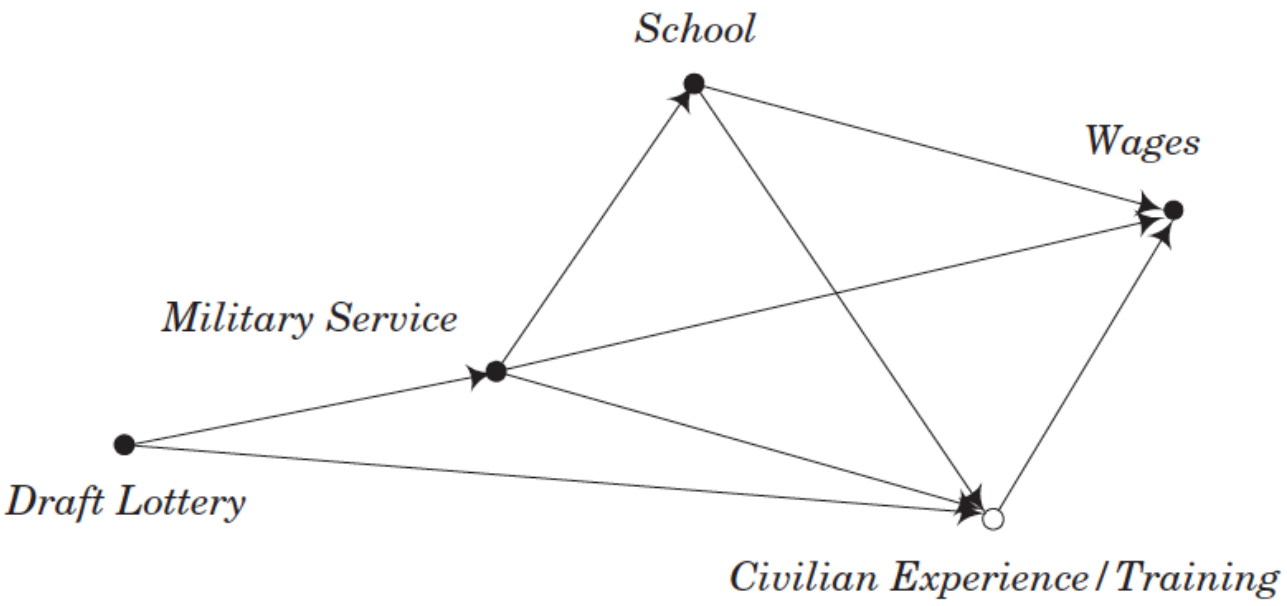

Figure 10.2 A directed graph for compliers with the Vietnam draft lottery as an IV for military service.

Two ways that an excellent estimate can be insufficiently deep.

Consider an optimistic scenario for which one obtains what all agree is a consistent estimate of the average treatment effect (ATE) for the effect of D on Y (as warranted, for example, by a consensus that one has conditioned on all variables that block all back-door paths from D to Y). As a result, a causal claim is valid by the standards of the counterfactual model, but

1. The ATE may not be a parameter of any fundamental interest.
2. The ATE does not entail any particular mechanism that explains how D brings about Y.

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A long tradition exists

For decades, social scientists have considered the explication of mechanisms through the introduction of intervening and mediating variables to be essential to sound explanatory practice in causal analysis. In *Socioeconomic Background and Achievement*:

> *much of the scientific quest is concerned with the search for intervening variables that will serve to interpret or explain gross associations presumed to reflect a causal relationship. (Duncan, Featherman, and Duncan 1972:12)*

The words "interpret" and "explain" are implicit references to the language of social research that is usually associated with Paul Lazarsfeld.

Kendall and Lazarsfeld

Kendall and Lazarsfeld considered two types of M-elaboration, which arise when the partial association between X and Y within strata defined by a "test factor" T is smaller than the original total association between X and Y.

1. If T follows X in time, then an M-type elaboration can be represented by a chain of mediation, $X \to T \to Y$ (Kendall and Lazarsfeld 1950:157). They refer to this type of elaboration as an *interpretation* of the association between X and Y, using a test factor that is an "intervening" variable.
2. If T precedes X in time, then an M-type elaboration can be represented by a fork of mutual dependence, $X \leftarrow T \to Y$ (Kendall and Lazarsfeld 1950:157). They refer to this type of elaboration as an *explanation* of the association between X and Y, using a test factor that is an "antecedent" variable.

Although Kendall and Lazarsfeld did not argue that causality can be established by what they call "interpretation," they were clearly interested in mechanistic accounts that reveal how causes bring about their effects. For example, they wrote:

> *When we interpret a result we try to determine the process through which the assumed cause is related to what we take to be its effect. How did the result come about? What are the "links" between the two variables? Answers to these questions are provided in the interpretation of the result. (Kendall and Lazarsfeld 1950:148)*

A long tradition of empirical social science exists that follows these basic ideas, in which a central goal of investigation is the search for variables – intervening, mediating, or processual variables – that can account for an association that is thought to be causal.

Pearl's Front-Door Criterion

Front-Door Criterion

If one or more unblocked back-door paths connect a causal variable to an outcome variable, the causal effect is identified by conditioning on a set of observed variables, $\{M\}$, that make up an identifying mechanism if:

- Condition 1 (exhaustiveness). The variables in the set $\{M\}$ intercept all directed paths from the causal variable to the outcome variable.
- Condition 2 (isolation). No unblocked back-door paths connect the causal variable to the variables in the set $\{M\}$, and all back-door paths from the variables in the set $\{M\}$ to the outcome variable can be blocked by conditioning on the causal variable.

A simple example of front-door identification

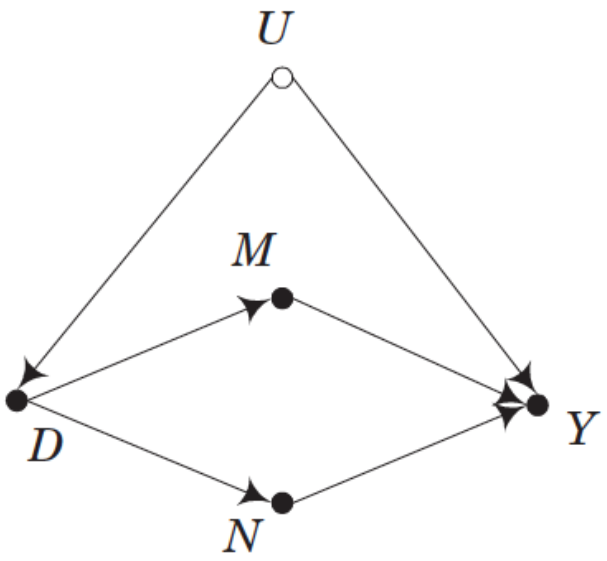


Figure 10.3 A directed graph in which M and N represent an exhaustive and isolated identifying mechanism for the causal effect of D on Y.

Just apply the back-door criterion ...

1. For $D \to M$ and $D \to N$, two back-door paths exist: $D \leftarrow U \to Y \leftarrow M$ and $D \leftarrow U \to Y \leftarrow N$, respectively. Because Y is a collider variable along both of these back-door paths, the causal effects $D \to M$ and $D \to N$ can be estimated consistently by their simple unconditional associations.
2. For $M \to Y$ and $N \to Y$, two back-door paths exist for each causal effect. For example, for $M \to Y$, they are $M \leftarrow D \leftarrow U \to Y$ and $M \leftarrow D \to N \to Y$. All four of the back-door paths can be blocked by conditioning on D. And, as a result, the casual effects $M \to Y$ and $N \to Y$ can be estimated consistently.

Assemble the estimate from its component pieces

For the ATE, the simplest case that is also consistent with the structure of Figure 10.3 is one where:

1. D is a binary variable
2. M, N, and Y are interval-scaled variables
3. all structural equations are linear and additive (e.g., $Y = f_Y(M, N, e_Y) = a_Y + b_M M + b_N N + e_Y$)
4. all individual-level causal effects do not vary in the levels of the variables in the mechanism

In this case, the ATE can be estimated consistently by a sum of products,

$$\left(\hat{b}_{D\to M} \times \hat{b}_{M\to Y}\right) + \left(\hat{b}_{D\to N} \times \hat{b}_{N\to Y}\right),$$

where

- $\hat{b}_{D\to M}$ is the estimated coefficient for M in a bivariate regression of Y on M,
- $\hat{b}_{D\to N}$ is the estimated coefficient for N in a bivariate regression of Y on N, and
- $\hat{b}_{M\to Y}$ and $\hat{b}_{N\to Y}$ are the estimated coefficients for M and N for a multiple regression of Y on M, N, and D.

Isolation is key

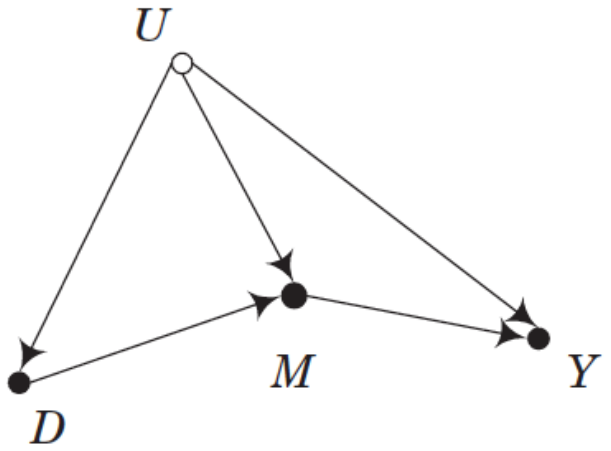
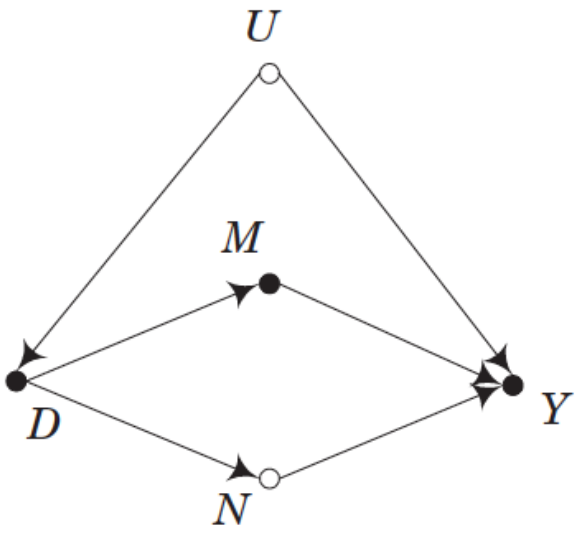


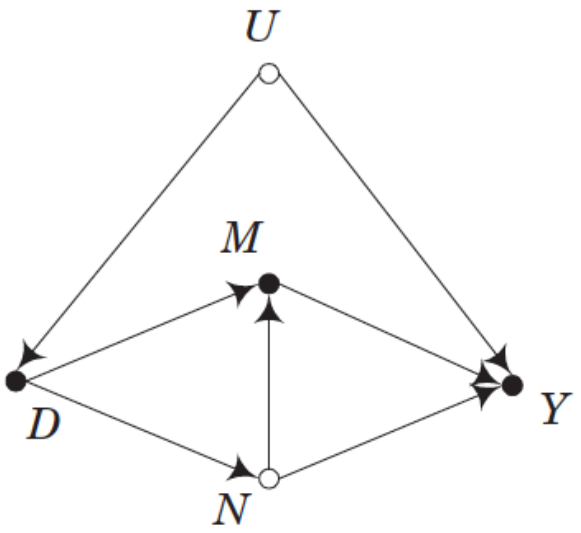
Figure 10.4 A directed graph in which M is not an isolated mechanism for the causal effect of D on Y.

The mechanistic variables must be isolated from otherwise unblocked back-door paths so that the back-door criterion can be applied in order to recover the full causal effect from the data.

Exhaustiveness as well



(a) The causal pathway via M is identified even though N is unobserved



(b) The causal pathway via M is not identified

Figure 10.5 Directed graphs in which one pathway in an exhaustive and isolated mechanism is unobserved.

Full and partial identification

1. If one wants to use a mechanism-based strategy to identify the full effect of a causal variable on an outcome variable, one must put forward an identifying mechanism that is exhaustive and isolated.
2. Isolation can be relaxed if the contamination can be adjusted away by conditioning.
3. The exhaustiveness requirement can be relaxed if one is satisfied with identifying only part of the causal effect of interest:
   - One must assume that the observed portion of the mechanism is not affected by any of the variables in the unobserved portion of the mechanism.
   - To assert this assumption, one typically needs to have a very specific theoretical model of the complete identifying mechanism, even though parts of it remain unobserved.

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A captivating call for theory construction *not* causal analysis

- Hedström and Swedberg (1998), *Social Mechanisms: An Analytical Approach to Social Theory.*
- Goldthorpe (2001), "Causation, Statistics, and Sociology"
- Hedström (2005), *Dissecting the Social: On the Principles of Analytical Sociology*
- Hedström and Bearman (2009), *The Oxford Handbook on Analytical Sociology*

Philosophy of science and explanation by mechanisms

- No need to resort to fringe philosophy of science (critical realism)
- Plain vanilla realism is fine
- The counterfactual modeling tradition in philosophy is robust
- A causal model variant (e.g., Hitchcock, Woodward) is well developed and provides a proper grounding

Adjudication between rival mechanisms

- We must go beyond theory construction, or else we will succumb to
  - mechanism anarchy
  - mechaminsm warlordism

Bottom line

"*How-possibly models* are often heuristically useful in constructing and exploring the space of possible mechanisms, but they are not adequate explanations. *How-actually models*, in contrast, describe real components, activities, and organizational features of the mechanism that in fact produces the phenomenon. Between these extremes is a range of *how-plausibly models* that are more or less consistent with the known constraints on the components, their activities, and their organization." (Craver 2007:111–12).

- The generative mechanisms movement is a call for the construction of how-possible and how-plausible models. Such activity is worthwhile in its own right, and it may help to inform empirical research.
- Causal analysis is a directed effort to evaluate the support for specific claims implied by how-actually models.

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Bottoming out

"Thinking About Mechanisms" (Machamer, Darden, and Craver 2000):

"... in a complete description of [a] mechanism, there are no gaps that leave specific steps unintelligible; the process as a whole is rendered intelligible in terms of entities and activities that are acceptable to a field at a time. ...
Nested hierarchical descriptions of mechanisms typically *bottom out* in lowest level mechanisms. These are the components that are accepted as relatively fundamental or taken to be unproblematic for the purposes of a given scientist, research group, or field. Bottoming out is relative: Different types of entities and activities are where a given field stops when constructing mechanisms. The explanation comes to an end, and description of lower-level mechanisms would be irrelevant to their interests." (Machamer et al. 2000:12–13)

Mechanism schema

- A *mechanism schema* is a representation of a mechanism in which some known details (or known nested levels) are suppressed for the sake of simplicity.
- Machamer et al. (2000:15) state, "a *mechanism schema* is a truncated abstract description of a mechanism that can be filled with descriptions of known component parts and activities."

Mechanism schema

In contrast, a *mechanism sketch* is quite different:

A sketch is an abstraction for which bottom out entities and activities cannot (yet) be supplied or which contains gaps in its stages. The productive continuity from one stage to the next has missing pieces, black boxes, which we do not yet know how to fill in. A sketch thus serves to indicate what further work needs to be done in order to have a mechanism schema. Sometimes a sketch has to be abandoned in the light of new findings. In other cases it may become a schema, serving as an abstraction that can be instantiated as needed. (Machamer et al. 2000:18)

Within this framework, one can conceive of causal analysis in the social sciences as the pursuit of explanations that bottom out.

One separable issue: What to do when causal states are coarse?

Three choices:

1. preserve the nominal causal states as originally defined and settle for a shallow account of how the cause produces its effect
2. preserve the nominal causal states as originally defined but then also attempt to deepen the shallow account by specifying a full mechanism for how the cause produces its effect
3. decompose the original nominal causal states into component causal states and then specify separable mechanisms that can account for the partial effects attributable to particular constitutive features, now redefined as lower-level nominal states

Choosing depends on

1. the state of relevant theory about the process under analysis
2. whether the constitutive features can be meaningfully separated and then measured
3. whether data with such measures are available
4. the standards that prevail in the field for which the study is intended

The pursuit of causal depth by investigation of nested mechanisms

- If a warranted claim of a counterfactually defined causal effect is properly regarded as insufficiently deep, the recourse is not to abandon the counterfactual model, but rather to investigate the nested mechanism that intercepts the effect of D on Y.
- Analysis may initially take the form of a set of alternative conjectured mechanisms.
- At some point, analysis must return to the particular observed relationship between D and Y.
- Empirical analysis of semigeneral mechanisms will lead to mechanism anarchy or mechanism warlordism.

Everyone would like be known as a deep theorist, but ... alas ...

- Although theoretical creativity and new measurement techniques may be required, opening up black boxes is, at least in part, an empirical pursuit.
- At some point, one must specify the causal states for the variables that constitute the mechanisms.
- Each link specified to give rise to an effect must be submitted to its own causal analysis.
- Lazarsfeld, Duncan, and all of their students recognized this ... and yet their perspective is almost completely absent from recent writing on mechanisms in sociology.